1. Introduction to DevSecOps

What is DevSecOps?

DevSecOps = Development + Security + Operations

DevSecOps is a software engineering approach that integrates security practices into every stage of the Software Development Life Cycle (SDLC).

Key Idea

→ Security is not treated as a separate final step before production.

→ Instead, security is integrated into every stage of the development process, from planning to monitoring.

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Main Objective

To build, test, release, and operate software securely, continuously, and efficiently.

DevSecOps in the SDLC

Shared Responsibility

Security becomes a shared responsibility among development, operations, and all other stakeholders, rather than being owned by a single team.

2. What is DevOps?

Definition

DevOps combines **Development + Operations** to deliver software faster, with better collaboration, automation, and reliability.

How it Works

Developers and operations teams work together throughout the software development lifecycle instead of in silos.

Same People.
Same Goal.
Faster Delivery.
Better Software.

Key Goals

Faster Delivery

Better Collaboration

Automation

Reliability

Focus Areas

Continuous Integration (CI)

Continuous Delivery (CD)

Infrastructure Automation

Monitoring

Rapid Feedback

DevOps Flow

Note on Security

Security may be handled separately or added later in some teams, but increasingly it is integrated into DevOps (e.g., DevSecOps).

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3. What is DevSecOps?

Definition

DevSecOps extends DevOps by embedding security into the whole lifecycle. Development Life Cycle (SDLC).

DevSecOps = DevOps + Continuous Security

Key Points

→ Security is a shared responsibility across developers, security teams, and operations.

→ Security is continuous, automated, and integrated into every stage of the development process.

→ The goal is to deliver secure, high-quality software faster, without compromising on security.

Common Practices

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Secure Coding

Follow secure coding guidelines and best practices.

Automated Scanning

Use automated tools to find vulnerabilities early.

Policy Checks

Enforce security policies and compliance rules.

Continuous Monitoring

Monitor for threats, risks, and anomalies in real-time.

DevSecOps Workflow

Result

By integrating security into every stage, DevSecOps helps organizations build and deliver secure, reliable, and compliant software at speed.

4. DevOps vs DevSecOps

DevOps

Focuses on speed, coliaboration and reliable delivery.

VS

DevSecOps

Builds on DevOps by integrating security into every stage.

Aspect	DevOps	DevSecOps
Focus	Speed, collaboration, reliability.	Speed + collaboration + reliability + security.
Security approach	May vary (often handled separately or later).	Embedded in every stage of the SDLC.
Testing	Functional and relegse-oriented testing.	Functional + security testing (SAST, DAST, etc.).
CI/CD pipeline	Build, test, and deploy automation.	Build, test, and deploy automation + security gates and policy checks.
Responsibility	Dev + Ops	Dev + Sec + Ops (shared responsibility).
Goal	Fast, reliable delivery of software.	Fast, reliable, secure delivery of software.

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Key Takeaway

DevSecOps does not replace DevOps — it strengthens it with continuous security.

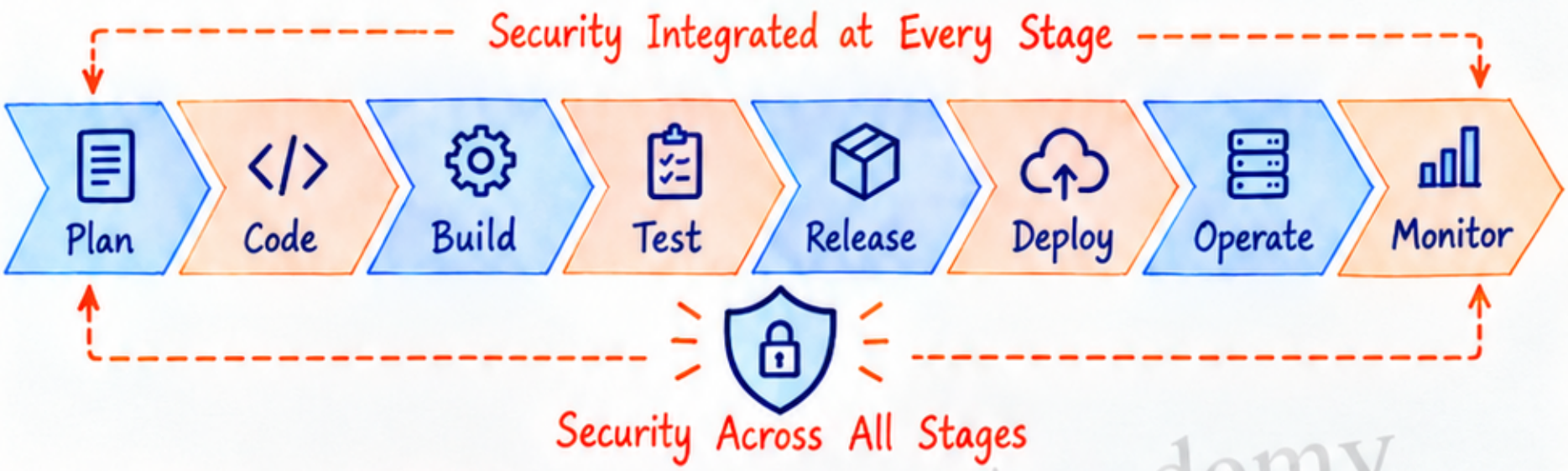
5. DevSecOps in the SDLC

What it Means

DevSecOps integrates security throughout the Software Development Life Cycle (SDLC), from planning to monitoring.

Security is built into every stage, rather than being added at the end.

DevSecOps Across the SDLC



Stage-wise Details

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1 Plan

Define security requirements.

2 Code

Follow secure coding practices.

3 Build

Run automated scans.

4 Test

Validate functionality and security.

5 Release

Verify approvals and risk status.

6 Deploy

Apply secure configurations.

7 Operate

Protect runtime environments.

8 Monitor

Detect incidents and vulnerabilities.

Key Takeaway

Security is continuous, not a one-time checkpoint.

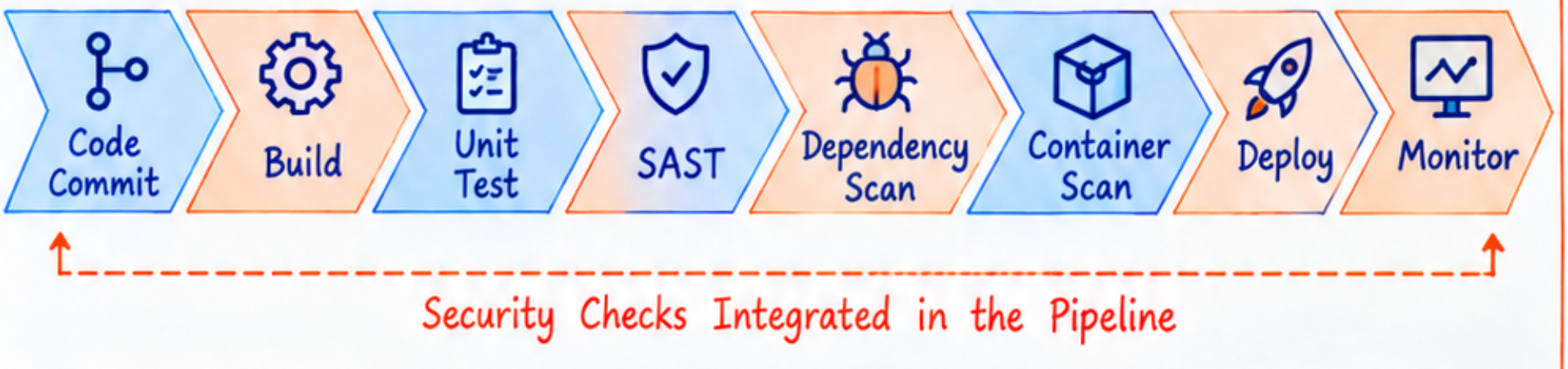
6. Secure CI/CD Pipeline

What is CI/CD?

CI/CD automates build, test, and deployment of software.

In DevSecOps, security checks are added inside the pipeline to catch issues early and deliver secure software.

CI/CD Pipeline with Security



Key Security Steps

1. **SAST (Static Application Security Testing)**
   Checks the source code for security issues like vulnerable functions, insecure configurations, etc.

2. **Dependency Scanning**
   Finds vulnerable third-party libraries and open source components used in the project.

3. **Container Scanning**
   Checks container images for known vulnerabilities before deployment.

4. **Policy Gates**
   Can stop the pipeline if critical security issues are found, preventing insecure releases from reaching production.

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Key Takeaway

Automation helps catch issues early and repeatedly, making the software more secure and reliable.

7. Real-World Example: Online Banking App

Example Application

An online banking system that allows customers to check balance, transfer money, pay bills, and manage their accounts through a web or mobile app.

Security Needs

Strong Authentication

Authorization (Access Control)

Encryption (Data in transit and at rest)

Secure Transactions

Audit Logging

Traditional Approach

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→ Focus is mainly on building features and delivering them quickly.

→ Security review may happen later, often just before production or after an incident.

DevSecOps Approach

1. **Define security requirements early.**
   Identify key security and compliance needs (e.g., data protection, regulatory requirements).
2. **Perform threat modeling.**
   Identify potential threats to the application and plan mitigations.
3. **Scan code and dependencies.**
   Use automated tools to find vulnerabilities in application code and third-party libraries.
4. **Scan containers and deployment configs.**
   Check container images, infrastructure as code, and configurations for security issues.
5. **Monitor production continuously.**
   Detect and respond to security threats, suspicious activity, and vulnerabilities in real time.

Goal

Protect customer data and financial transactions.

8. Common DevSecOps Tools

Popular tools used across different stages of the DevSecOps pipeline:

Source Code
Version control and code collaboration.

Git | GitHub | GitLab

CI/CD
Automate build, test and deployment.

Jenkins | GitHub Actions | GitLab CI

SAST
Static code analysis to find security issues early.

Semgrep | SonarQube

Dependency Scanning
Identify vulnerable open-source dependencies.

Snyk | OWASP Dependency-Check

Secret Scanning
Detect hardcoded secrets in code and repositories.

Gitleaks

Container Scanning
Scan container images for vulnerabilities.

Trivy | Grype

IaC Scanning
Find misconfigurations in infrastructure as code.

Checkov | Trivy

Dynamic Testing
Test running applications for security vulnerabilities.

OWASP ZAP

Monitoring / Runtime
Monitor for threats, anomalies and runtime security.

Falco | Prometheus | Grafana

Tools support the process, but **secure culture** and **automation** are equally important.

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9. Benefits of DevSecOps

Why DevSecOps?

DevSecOps brings security into every stage of the software development lifecycle, helping organizations build and deliver software that is more secure, reliable, and resilient.

Find problems early, fix them faster, and release with more confidence.

Key Benefits

1. Early vulnerability detection – find and fix security issues in early stages.
2. Lower cost of fixing security issues – cheaper to fix problems early than after release.
3. Faster and safer releases – deliver features quickly without compromising security.
4. Automated, repeatable security checks – consistent and reliable security testing in the pipeline.
5. Better compliance and audit readiness – meet regulatory requirements with proper tracking and evidence.
6. Improved collaboration across teams – development, security, and operations work together.
7. Continuous protection in production – monitor, detect, and respond to threats in real time.
8. Reduced risk to applications and data – keep systems and data safe from security threats.

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DevSecOps Benefits at a Glance

10. Shift Left, Shift Right & Conclusion

Shift Left

Shift Left means introducing security early in the software development process, during planning, design, coding, and testing.

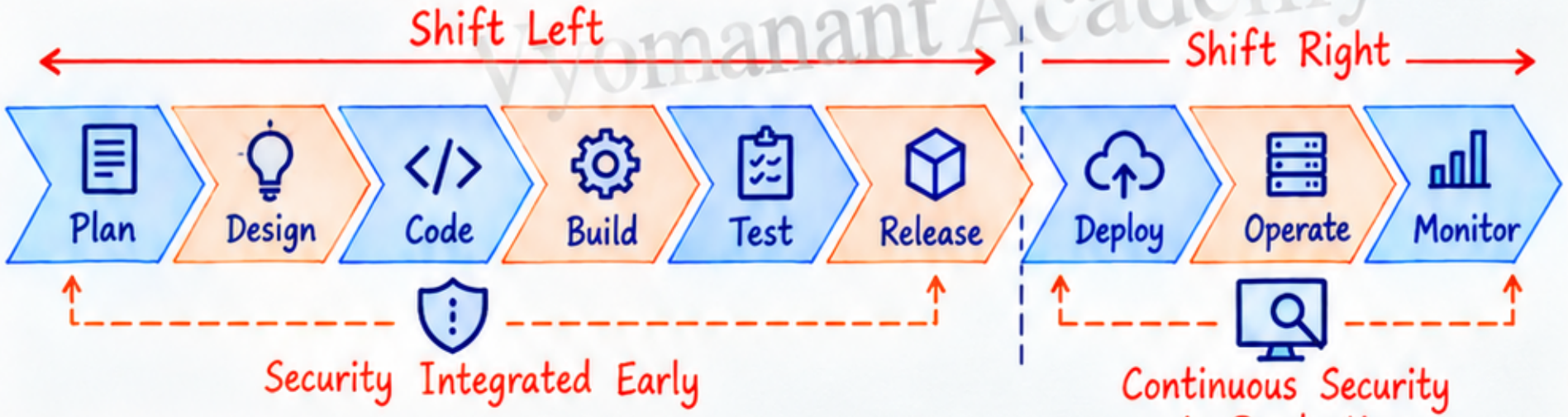
Example: Detect hardcoded secrets, vulnerable dependencies, or insecure code before the code is merged.

Shift Right

Shift Right means continuing security in production through monitoring, detection, and response.

Example: Detect suspicious runtime activity, unusual behavior, or attacks in the production environment.

End-to-End View: Shift Left to Shift Right



Conclusion

1. DevOps improves speed and reliability.
2. DevSecOps adds continuous security integration.
3. Security becomes a shared responsibility among development, operations, and all other stakeholders.
4. Goal: deliver software fast, reliably, and securely.

People + Process + Technology = Secure Software

DevSecOps = Secure software delivery as a continuous practice.

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